Code No: 117EE

R13

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, February/March - 2022
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer Any Five Questions
All Questions carry equal marks
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1.a) How to implement security by file permissions? Explain the utilities used to implement security at various levels in the system and at file access level.
b) Write a shell script to read a file using command line arguments and print the number of words, lines in the file. [8+7]

2.a) List and explain various system commands in awk with examples for each.
b) Write an awk script to print series of fibonacci series in between a given range. [7+8]

3.a) Discuss the relative merits and demerits of file locking and record locking.
b) Give the syntax of fcntl() system call and explain briefly. [8+7]

4.a) Differentiate between link() and symlink() system calls.
b) Explain various functions used in listing the content of a directory with clear syntax and sample code. [8+7]

5.a) What are the macros to examine the termination status returned by wait and waitpid? Demonstrate with an example the various exit status.
b) Differentiate between wait() and waitpid() system calls. [9+6]

6.a) What are the properties of the parent that are inherited by the child process?
b) Illustrate kill() and raise() system calls with an example program in C. [6+9]

7.a) Write a brief note on identifiers and keys used in system V IPC.
b) What happens if the cmdstring executed by popen( ) with a type of "r" write to its standard error? Write a program to illustrate popen( ). [6+9]

8. Write and explain a client-server program for inter process communication using shared memory IPC mechanism. [15]

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